

Application Note 09 - Hot Water and Chemical Service

How to distinguish pump limits, seal limits and chemistry review

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Core principle

The allowable pump-body temperature and the allowable seal temperature are separate constraints. The lower applicable limit governs the standard configuration.

A pump may be mechanically rated for a process temperature that exceeds the continuous rating of its standard seal. In such cases, an optional seal may be required even though the pump model itself remains suitable.

Worked examples

Case	Pump-body check	Seal check	Preliminary conclusion
HX-240 at 95°C	Within 95°C pump limit	EPDM 90°C limit exceeded	Use FKM; review chemistry
HX-300 at 96°C	Within 110°C pump limit	Standard FKM within limit	Standard seal acceptable subject to chemistry
HX-400 at 140°C	At pump continuous limit	FKM 130°C limit exceeded	Use PTFE; engineering review
HX-520 at 132°C	Within 135°C pump limit	Hygienic EPDM 125°C exceeded	Use hygienic FKM

Customer-response guidance

When answering technical questions, cite both the model-specific operating limit and the relevant seal-material guidance. If the chemistry is unknown or aggressive, state that final compatibility remains subject to review.